

1. Project Overview

The project focuses on analyzing one year of pizza sales data to understand business performance and customer purchasing trends. Using SQL for data extraction and Tableau for visualization, the goal was to identify patterns in revenue, order frequency, product performance, and customer preferences.

2. Objectives

1. Calculate the key business KPIs such as:
 - Total Revenue
 - Average Order Value
 - Total Pizzas Sold
 - Total Orders
 - Average Pizzas per Order
2. Build interactive visual dashboards to uncover:
 - Hourly & Weekly order trends
 - Sales distribution by category and size
 - Best & worst-selling pizzas
 - Customer demand patterns

3. Tools & Techniques Used

SQL

Used for:

- Data cleaning & preprocessing
- Calculating KPIs
- Group-wise aggregations
- Identifying top/bottom performers

Tableau

Used for:

- Building dynamic dashboards
- Using slicers & filters (month, category, date range)
- Visual storytelling for business insights

PIZZA SALES REPORT

Problem Statement:-

KPI'S REQUIREMENT

We need to analyze key indicators for our pizza sales data to gain insights into our business performance. Specifically, we want to calculate the following metrics:

- Total Revenue: The sum of the total price of all pizza orders.
- Average Order Value: The average amount spent per order, calculated by dividing the total revenue by the total number of orders.
- Total Pizzas Sold: The sum of the quantities of all pizzas sold.
- Total Orders: The total number of orders placed.
- Average Pizzas Per Order: The average number of pizzas sold per order, calculated by dividing the total number of pizzas sold by the total number of orders.

CHARTS REQUIREMENT

We would like to visualize various aspects of our pizza sales data to gain insights and understand key trends. We have identified the following requirements for creating charts:

1. Hourly Trend for Total Orders:

- Create a stacked bar chart that displays the hourly trend of total orders over a specific time period.
- This chart will help us identify any patterns or fluctuations in order volumes on a hourly basis.

2. Weekly Trend for Total Orders:

- Create a line chart that illustrates the weekly trend of total orders throughout the year.
- This chart will allow us to identify peak weeks or periods of high order activity.

3. Percentage of Sales by Pizza Category:

- Create a pie chart that shows the distribution of sales across different pizza categories.
- This chart will provide insights into the popularity of various pizza categories and their contribution to overall sales.

4. Percentage of Sales by Pizza Size:

- Generate a pie chart that represents the percentage of sales attributed to different pizza sizes.
- This chart will help us understand customer preferences for pizza sizes and their impact on sales.

5. Total Pizzas Sold by Pizza Category:

- Create a funnel chart that presents the total number of pizzas sold for each pizza category.
- This chart will allow us to compare the sales performance of different pizza categories.

6. Top 5 Best Sellers by Revenue, Total Quantity and Total Orders:

- Create a bar chart highlighting the top 5 best-selling pizzas based on the Revenue, Total Quantity, and Total Orders.
- This chart will help us identify the most popular pizza options.

7. Bottom 5 Best Sellers by Revenue, Total Quantity and Total Orders:

- Create a bar chart showcasing the bottom 5 worst-selling pizzas based on the Revenue, Total Quantity, and Total Orders.
- This chart will enable us to identify underperforming or less popular pizza options.

PIZZA SALES SQL QUERIES

A. KPI's

1. Total Revenue:

```
select Sum(total_price) as Total_revenue from pizza_sales;
```

	Total_revenue
▶	817860.0499999993

2. Average Order Value

```
select sum(total_price)/ count(distinct order_id) as Avg_order_Value from pizza_sales;
```

	Avg_order_Value
▶	38.307262295081635

3. Total Pizzas Sold

```
select sum(quantity) as Total_Pizza_Sold from pizza_sales;
```

	Total_Pizza_Sold
▶	49574

4. Total Orders

```
select count(distinct order_id) as Total_orders from pizza_sales;
```

	Total_orders
▶	21350

5. Average Pizzas Per Order

```
select sum(quantity)/ count(distinct order_id) as Avg_pizza_per_order from pizza_sales;
```

	Avg_pizza_per_order
	2.3220

B. Hourly Trend for Total Pizzas Sold

```
select hour(order_time) as order_hour,  
sum(quantity) as Total_pizza_sold  
from pizza_sales  
group by Hour(order_time)  
order by order_hour;
```

order_hour	Total_pizza_sold
9	4
10	18
11	2728
12	6776
13	6413
14	3613
15	3216
16	4239
17	5211
18	5417
19	4406
20	3534
21	2545
22	1386
23	68

C. Weekly Trend for Orders

```
select week(order_date,3) as Week_Number, Year(order_date) as Order_year,  
count(distinct order_id) as Total_orders  
from pizza_sales  
group by Week_Number, Order_year  
order by Order_year, Week_Number;
```

Week_Number	Order_year	Total_orders
1	2015	254
2	2015	427
3	2015	400
4	2015	415
5	2015	436
6	2015	422
7	2015	423
8	2015	393
9	2015	409
10	2015	420
11	2015	404
12	2015	416
13	2015	427
14	2015	433
15	2015	408
16	2015	414
17	2015	437
18	2015	423
19	2015	399
20	2015	458
21	2015	414
22	2015	390
23	2015	423
24	2015	418
25	2015	410

Week_Number	Order_year	Total_orders
26	2015	416
27	2015	474
28	2015	417
29	2015	420
30	2015	433
31	2015	419
32	2015	426
33	2015	435
34	2015	407
35	2015	394
36	2015	397
37	2015	435
38	2015	423
39	2015	288
40	2015	433
41	2015	334
42	2015	386
43	2015	352
44	2015	371
45	2015	394
46	2015	400
47	2015	392
48	2015	491
49	2015	424
50	2015	417

D. % of Sales by Pizza Category

```
select pizza_category, sum(total_price) as Total_sales,
sum(total_Price) *100 / (select sum(total_price) from pizza_sales) as PCT
from pizza_sales
group by pizza_category;
```

pizza_category	PCT
Classic	26.9059602556699
Veggie	23.682590927384783
Supreme	25.45631126009884
Chicken	23.955137556847493

MONTH SPECIFIC

```
select pizza_category, sum(total_price) as Total_sales, sum(total_Price)
*100 / (select sum(total_price) from pizza_sales) as PCT
from pizza_sales
where month(order_date) = 1
group by pizza_category;
```

pizza_category	Total_sales	PCT
Classic	18619.4	2.276599767894295
Veggie	17055.400000000027	2.0853690065922863
Supreme	17929.749999999996	2.1922760501628793
Chicken	16188.75	1.9794034443912671

E. % of Sales by Pizza Size

```
select pizza_size, cast(sum(total_price) as decimal(10,2)) as
Total_revenue,
cast(sum(total_price) *100/ (select sum(total_price) from pizza_sales) as
decimal(10,2)) as PCT
from pizza_sales
group by pizza_size
order by PCT desc;
```

pizza_size	Total_revenue	PCT
L	375318.70	45.89
M	249382.25	30.49
S	178076.50	21.77
XL	14076.00	1.72
XXL	1006.60	0.12

F. Total Pizzas Sold by Pizza Category

```
select pizza_category, sum(quantity) as Total_pizza_sold
from pizza_sales
group by pizza_category;
```

pizza_category	Total_pizza_sold
Classic	14888
Veggie	11649
Supreme	11987
Chicken	11050

MONTH SPECIFIC

```
select pizza_category, sum(quantity) as Total_pizza_sold
from pizza_sales
where month(order_date)=2
group by pizza_category
order by Total_pizza_sold desc;
```

pizza_category	Total_pizza_sold
Classic	1178
Supreme	964
Veggie	944
Chicken	875

G. Top 5 Pizzas by Revenue

```
select pizza_name, sum(total_price) as Total_Revenue from pizza_sales
group by pizza_name
order by Total_revenue desc
limit 5;
```

pizza_name	Total_Revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5
The Classic Deluxe Pizza	38180.5
The Spicy Italian Pizza	34831.25

H. Bottom 5 Pizzas by Revenue

```
select pizza_name, sum(total_price) as Total_Revenue from pizza_sales
group by pizza_name
order by Total_revenue asc
limit 5;
```

pizza_name	Total_Revenue
The Brie Carre Pizza	11588.4999999999
The Green Garden Pizza	13955.75
The Spinach Supreme Pizza	15277.75
The Mediterranean Pizza	15360.5
The Spinach Pesto Pizza	15596

I. Top 5 Pizzas by Quantity

```
select pizza_name, sum(quantity) as Total_Quantity from pizza_sales
group by pizza_name
order by Total_Quantity desc
limit 5;
```

pizza_name	Total_Quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

J. Bottom 5 Pizzas by Quantity

```
select pizza_name, sum(quantity) as Total_Quantity from pizza_sales
group by pizza_name
order by Total_Quantity asc
limit 5;
```

pizza_name	Total_Quantity
The Brie Carre Pizza	490
The Mediterranean Pizza	934
The Calabrese Pizza	937
The Spinach Supreme Pizza	950
The Soppressata Pizza	961

K. Top 5 Pizzas by Total Orders

```
select pizza_name, count(distinct order_id) as Total_order
from pizza_sales
group by pizza_name
order by Total_order Desc
limit 5;
```

pizza_name	Total_order
The Classic Deluxe Pizza	2329
The Hawaiian Pizza	2280
The Pepperoni Pizza	2278
The Barbecue Chicken Pizza	2273
The Thai Chicken Pizza	2225

L. Bottom 5 Pizzas by Total Orders

```
select pizza_name, count(distinct order_id) as Total_order
from pizza_sales
group by pizza_name
order by Total_order asc
limit 5;
```

pizza_name	Total_order
The Brie Carre Pizza	480
The Mediterranean Pizza	912
The Calabrese Pizza	918
The Spinach Supreme Pizza	918
The Chicken Pesto Pizza	938